

Pillar 1: Narrative Answers

- **1. Software Process:** A structured set of activities ensuring uniform and compatible software across different sites.
- **2. Strategic Assessment:** Evaluating if the project aligns with the company's mission, like shifting to "Predictive Medicine."
- **3. Project vs. Programme:** A project is a single effort, while a programme is a collection of related projects.
- **4. Technical Assessment:** Evaluating if we have the skills/hardware; risks include lacking training data or GPU power.
- **5. Time Value of Money:** Money loses value over time due to inflation and missed investment opportunities.
- **6. J-Curve:** A curve showing negative cash flow during development, hitting maximum debt just before launch.
- **7. Benefits:** Direct benefits are financial (reduced staffing costs), while intangible benefits are non-financial (brand reputation).
- **8. Waterfall:** Sequential model. feedback is late because users only see the system in the final testing phase, making errors expensive to fix.
- **9. V-Model:** A symmetrical development and testing model, excellent for compliance (like GDPR) because tests align with requirements.
- **10. Prototyping:** "Throwaway" is a temporary mockup for quick feedback, while "Evolutionary" is a continuously improved working model.
- **11. Spiral Model:** A risk-driven iterative model that helps decide whether to continue or abort before spending the full budget.
- **12. Agile/Scrum:** Uses short "Sprints" for quick testing and immediate feedback in changing environments.
- **13. Sync vs. Async:** Synchronous means working simultaneously; asynchronous involves time gaps and requires strict documentation.
- **14. Follow-the-Sun:** Passing work across time zones to achieve continuous 16-24 hour productivity.

- 15. **Bespoke vs. COTS:** COTS is ready-made for standard tasks (billing), while Bespoke is custom-built for a unique competitive advantage (AI).
- 16. Step 3 (Characteristics): Analyzes project risks and requirement clarity to dictate the right approach (e.g., Agile or Spiral).
- 17. Incremental Delivery: Releasing project parts in stages (like Login first) to encourage early user adoption.
- 18. Step-wise: An iterative process where cost estimations might require revisiting the initial scope.
- 19. Organizational Infrastructure: Defines the reporting structure so teams know exactly who approves what.
- 20. Hybrid Lifecycle: Combining different models (e.g., Waterfall for security, Agile for UI) to balance safety with speed.

Pillar 2: Case Answers

- 1. J-Curve & Healthcare: Healthcare complexity causes deep negative cash flow; returns only appear after full rollout.
- **2. Lifecycle Choice: Spiral.** It is best for high-risk projects to test hypotheses before full implementation.
- **3. Prototype Choice: Throwaway.** Mockups manage expectations quickly without committing to complex code.
- 4. Hybrid Solution: Use **Agile** for UI speed and the **V-Model** for strict security audit trails.
- 5. Technical Feasibility: Weak infrastructure is a major risk; a hardware upgrade must be budgeted.
- 6. Buy vs. Build Risks: Buying risks include vendor lock-in and loss of Intellectual Property (IP).
- 7. Social Impact: Job losses cause stakeholder resistance, requiring Change Management and staff retraining.
- 8. Synchronization: Use fixed milestones so Agile teams sync perfectly with V-Model integration dates.
- **9. Audit Trail: V-Model.** It provides a clear matrix matching every requirement to a specific test.

- **10. Outsourcing Decision: No.** A 40% risk increase in healthcare outweighs the cost savings.
- 11. Interdependence: Projects in a programme are linked; if the AI fails, the database becomes useless.
- 12. Async Management: Use 24/7 shared dashboards (like Jira/Wiki) to reduce the need for live meetings.
- 13. The "Black Art" Risk: Standard processes ensure *any* developer can fix bugs, avoiding reliance on one "brilliant" person.
- 14. Delivery Risk: "Big Bang" has a high crash risk, while "Incremental" contains errors to a small pilot area.
- 15. Tie-breaker: Choose the project that best aligns with the company's long-term strategic mission.
- 16. V-Model Fix: Return to the "Requirements Definition" phase (the opposite side of the 'V').
- 17. Infrastructure Issue: Cultural conflicts disrupt communication, which delays project handovers.
- 18. Expectation Gap: Shift the stakeholder's focus to reviewing major features monthly instead of reviewing code bi-weekly.
- 19. Competitor Launch: **Pivot** to add a unique feature; exiting wastes the money already spent (sunk cost).
- 20. Compliance Cost: Include it as a "Non-Functional Requirement" cost in the budget.

Pillar 3: Quantitative Key (Formulas & Results)

- 1. Annual Net Benefit = \$4M.
- 2. ROI = 133.3%.
- 3. Payback Period = Year 3.
- 4. NPV = \$0.678M.
- 5. Decision = **Go-Go** (Positive NPV means the project creates value).